Kodeguru Tech Solutions

CLIENT CASE STUDY

Read online: kodeguru-portfolio.netlify.app/work/cream

ENTERTAINMENT AND CREATOR ECONOMY · NIGERIA

How we helped CREAM stop losing customers to downtime and keep more of every sale

CREAM is a platform where African musicians and creators sell their work and fans win prizes for supporting them. When CREAM came to us, the platform went offline once or twice a week, was hard to change, and earned most of its money through a channel that gave a large share away. This is what we changed and what it did for the business.

0 weekly outages now, down from 1 to 2 every week	**~35k** unique users a day across CREAM's tracked channels	**4** ways to pay: airtime, CREAM tokens, card and bank transfer
90% less mobile data needed to open a song page, after this month's tune-up		

THE CLIENT

Who CREAM is

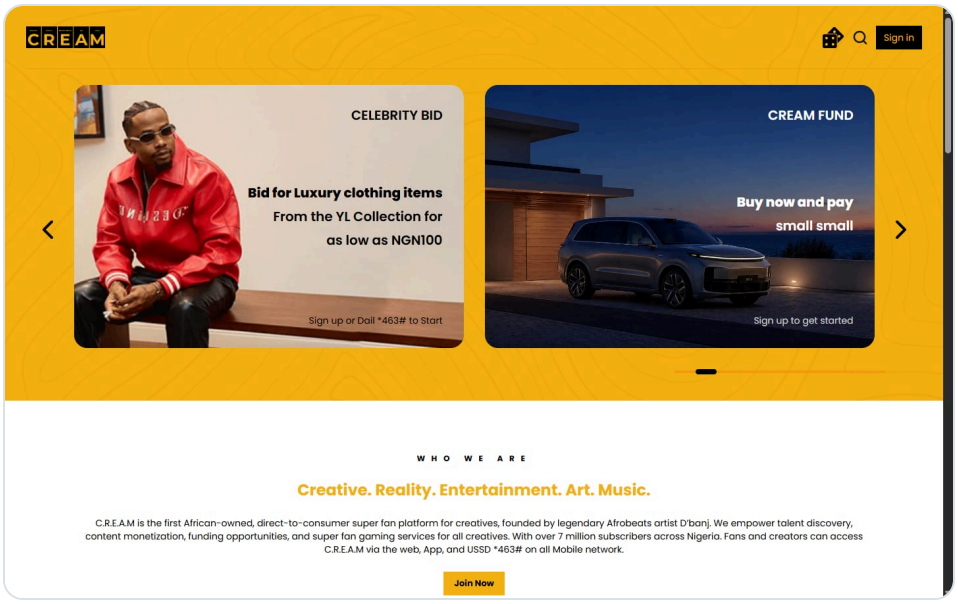
CREAM (Creative, Reality, Entertainment, Art, Music) describes itself as Africa's first African-owned, direct-to-consumer super-fan platform. It was founded by Afrobeats artist

D'banj and reports over 7 million subscribers across Nigeria.

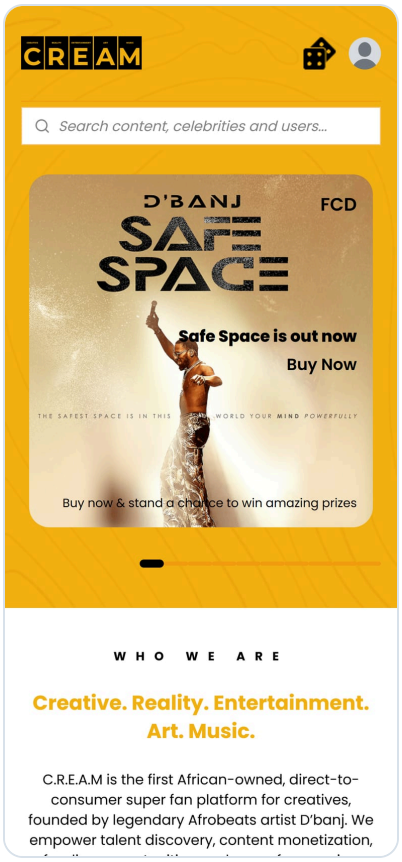
The business has two kinds of customer:

- **Creators** (musicians, video makers, comedians) who upload their work, sell it and get paid.
- **Fans** who stream, buy and download that work, and who take part in prize draws, celebrity auctions and talent contests.

Fans reach CREAM in three ways: the website at creamplatform.com, the rewards site, and by dialling ***463#** on any MTN phone.



creamplatform.com today, on a laptop.



The same page on a phone.

WHERE THINGS STOOD

The problems CREAM brought to us

None of these problems were about one bug. Each one cost the business money or customers every week.

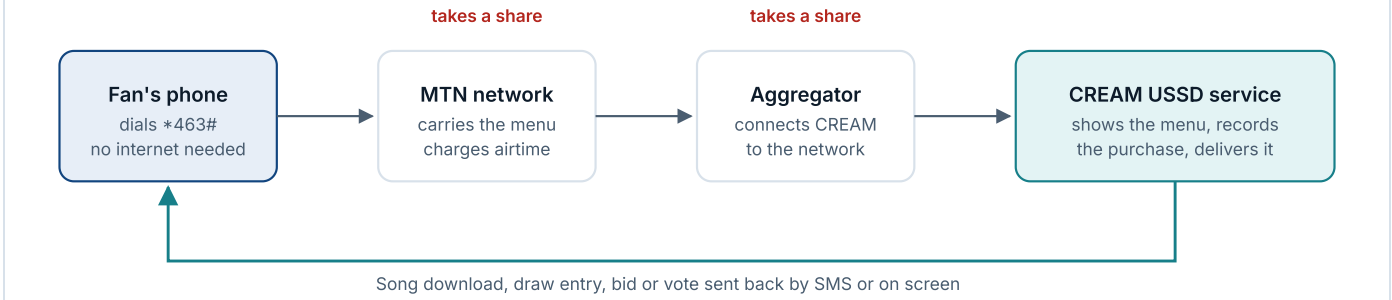
PROBLEM	WHAT IT COST THE BUSINESS
The site went down once or twice a week. The server was not set up to manage its resources, so busy periods knocked it over.	Every outage meant fans who could not buy, vote or enter a draw, and creators who could not sell. Some of those people did not come back.
Everything was tangled together. The website, the business rules and the data all lived in one block of old PHP code. Changes were made directly on the live server.	Adding a feature or fixing a bug was slow and risky. One change could break an unrelated part of the site.
It worked badly on phones. The layout broke on small screens.	Most CREAM fans use phones, so most visitors had a poor experience.
Google could not read it. Song and creator pages were invisible to search engines.	A content business depends on people finding its content. CREAM was missing free traffic from search.
Most sales came through airtime. Fans paid by dialling *463# and paying with phone credit.	Airtime payments are split with the mobile network and the company that connects CREAM to it, so CREAM kept a smaller share of each sale.
Every app had its own login. CREAM runs at least four apps, and each one asked fans to sign in separately.	Fans moving from the main site to rewards or CreamPay had to sign in again, and some gave up. Each app also kept its own idea of who a customer was.
Paying people was expensive. Staff, contest winners and artists were paid through outside banks.	Every payout carried bank charges, and money moved slowly.

PLAIN-LANGUAGE EXPLAINER

Why a business like CREAM needs USSD

USSD is the menu that appears when you dial a code like ***463#**. It looks old-fashioned, but it solves a problem a website cannot: it works on **any phone**, including basic phones, with **no internet and no data bundle**. In Nigeria and across Africa, that means millions of people who would never open a website or install an app can still buy from you in a few seconds.

The customer pays with airtime, which they already have. That convenience has a cost: the mobile network and the aggregator (the company that connects a business to the network) each take a share of the payment.



A USSD sale reaches people with no internet, but the network and the aggregator each take a share before CREAM is paid.

WHAT WE DID

Nine changes, each tied to a business result

Writing code is only part of the job. The real value is in understanding where a business is losing money or customers, deciding what to build, and checking that it works. Each change below starts from a business problem.

Stable servers that stay up

PROBLEM

The platform went offline once or twice a week because the server could not manage its load.

WHAT WE BUILT

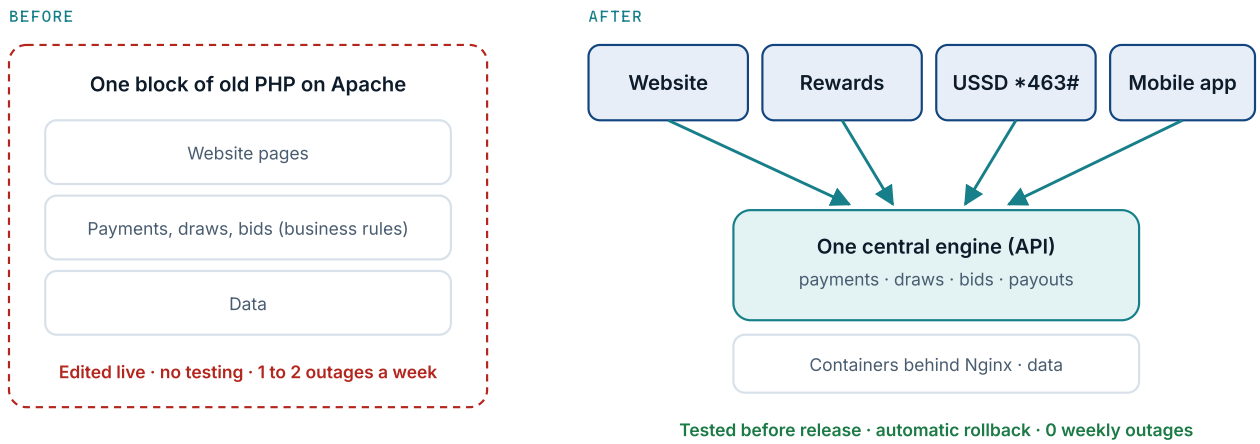
We packaged the back end into containers, so each part runs in its own controlled space with the resources it needs. We replaced the old Apache web server with Nginx, which handles heavy traffic with far less strain. New versions now go live through an automatic process that checks the new version is healthy and switches back to the last good version on its own if it is not.

RESULT

No more weekly outages. An update now interrupts the site for under half a second, and a bad release can no longer take the site down.

One shared engine for every channel

PROBLEM	The website, the business rules and the data were one tangled block. Every change risked breaking something else.
WHAT WE BUILT	We separated the "front" (what customers see) from the "engine" (the rules about payments, draws, bids and payouts). The engine is now one central service, and the website, the rewards site, the USSD menu and the mobile app all use it.
RESULT	A rule changes in one place and every channel gets it at once. New products, like a new contest or payment method, launch in days instead of weeks.



Before and after. Every customer channel now uses the same engine, so rules change once and apply everywhere.

One login for every CREAM app

PROBLEM

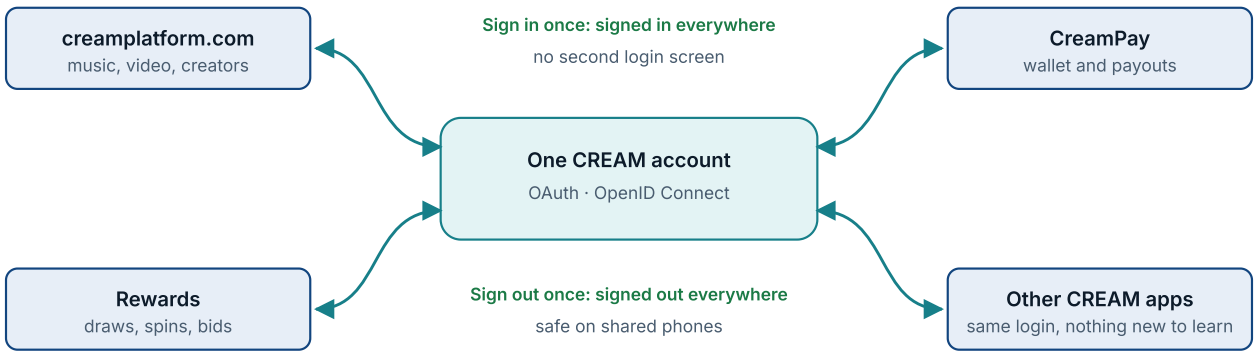
CREAM runs at least four apps, including the main site, the rewards site and CreamPay. Each had its own login, so a fan signed in to one was a stranger to the others.

WHAT WE BUILT

A single CREAM account system that every app trusts, built on OAuth and OpenID Connect, the same open standards behind "Sign in with Google". Sign in on any CREAM app and you are signed in on all of them, without seeing another login screen. Sign out on one and you are signed out everywhere, which matters on shared phones.

RESULT

Fans move between buying music, playing for prizes and managing their wallet without friction. CREAM sees one customer across every app, so a song bought on the main site can be matched to a draw entry on the rewards site, and new apps can join the same login in days.



Every CREAM app checks the same account, so one sign-in or sign-out applies to all of them.

A website that works on every phone and shows up on Google

PROBLEM

The site broke on small screens, and song and creator pages could not be found on Google.

WHAT WE BUILT

We rebuilt the website from scratch so it adapts to any screen. We then moved it to a set-up (Next.js) that prepares each page in full before it reaches the visitor, which is what search engines need in order to read it. Every song, video and creator page now has its own title, description and share preview, and a site map lists about 2,700 content pages for Google. The site is available in English, Pidgin, Yoruba, Igbo, Hausa and French.

RESULT

Fans can find CREAM's content through search, and links shared on WhatsApp and social media show a proper preview card. That brings in visitors without paid advertising.

A faster, safer USSD service

PROBLEM

The *463# menu ran on loose PHP scripts that were hard to change and test. The song list took over 10 seconds to load, long enough for the session to time out and the sale to be lost.

WHAT WE BUILT

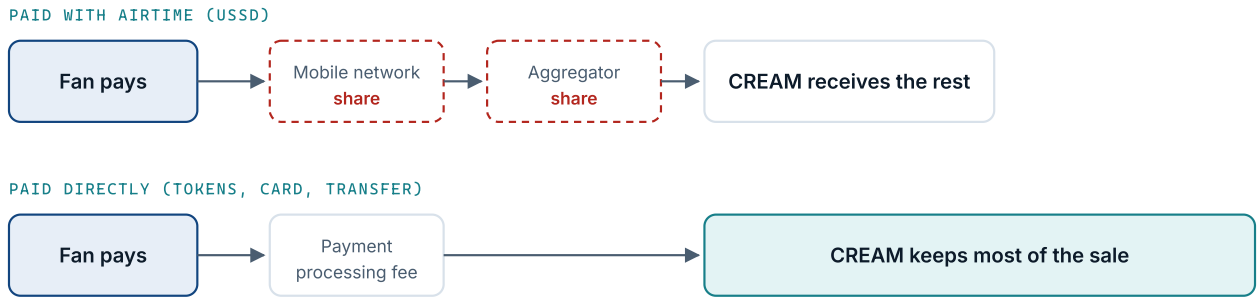
We rebuilt the USSD service as one organised application that answers MTN exactly as before, so MTN did not have to change anything. It records every purchase before charging, never charges twice for the same request, and ignores duplicate payment confirmations. The song list now loads instantly from a copy that refreshes every five minutes. A phone simulator and about 190 automated checks let the team test every menu before customers see it.

RESULT

Fans are not double-charged, nobody pays without receiving what they bought, and fewer sales are lost to timeouts. New games can be tested on staff phones before they go public.

Direct payments that keep more of each sale

PROBLEM	Most revenue came through USSD airtime, which is shared with the mobile network and the aggregator.
WHAT WE BUILT	Fans can now buy on the website and app with CREAM tokens, a debit card or a bank transfer (through Paystack, with OPay being rolled out). USSD stays available for fans without internet.
RESULT	Every sale that moves from airtime to a direct payment leaves more money with CREAM, because there is no network or aggregator share to pay.



The same sale, two routes. The diagram shows who takes a share; the exact percentages depend on CREAM's agreements.

CreamPay: CREAM’s own digital bank

PROBLEM	CREAM paid staff, contest winners and artists through outside banks, and paid bank charges on every transfer.
WHAT WE BUILT	CreamPay (pay.creamplatform.com), a wallet for creators and businesses. Users send, spend and receive money in naira, US dollars, euros and pounds, with virtual cards and instant transfers. CREAM now runs its own payouts through it: staff salaries, Best of the Streets winners, artist earnings and prize-draw winners.
RESULT	Most of the bank fees on payouts are gone, money reaches people faster, and winners and artists stay inside the CREAM world, where they can spend their earnings on the platform.



Every payout now starts from CREAM's own wallet instead of an outside bank.

Rewards that make every purchase worth more

PROBLEM

Buying a song was a one-off transaction, with little reason to come back the next day.

WHAT WE BUILT

Buying a song now enters the fan into a cash prize draw automatically. The rewards site (rewards.creamplatform.com) adds Koko Draw, Spin D' Koko, and lowest-unique-bid auctions for cash and celebrity items. In those auctions the winner is the lowest bid nobody else placed: if two fans both bid 1, then 1 is no longer unique and 2 becomes the lowest unique bid. Best of the Streets lets talent register, upload a performance video and collect public votes, with a live results screen at the watch-party venue.

RESULT

Fans have a reason to come back daily, every purchase supports a creator, and CREAM sells more content because of the prizes attached.

A professional way of working

PROBLEM

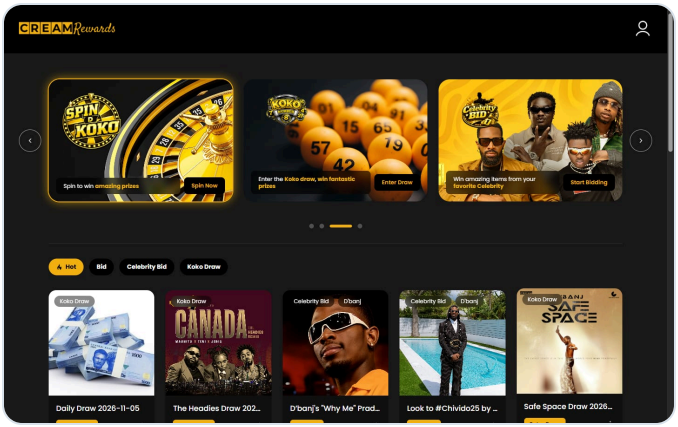
There was no company code repository and no review step. Changes went straight onto the live server.

WHAT WE BUILT

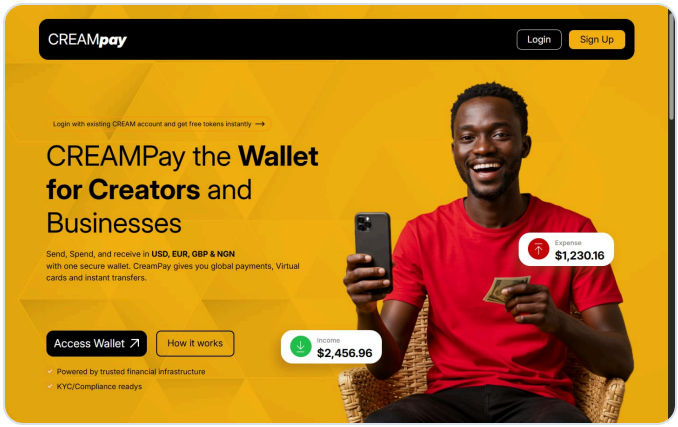
A company GitHub account where every change is reviewed and automatically checked before it goes live, with a message posted to the team's Slack after each release. We also added user analytics (Amplitude), so CREAM can see which features fans use and where purchases drop off.

RESULT

CREAM owns its code, can bring in new developers safely, and makes product decisions from real usage data instead of guesses.



rewards.creamplatform.com: draws, spins and celebrity auctions.

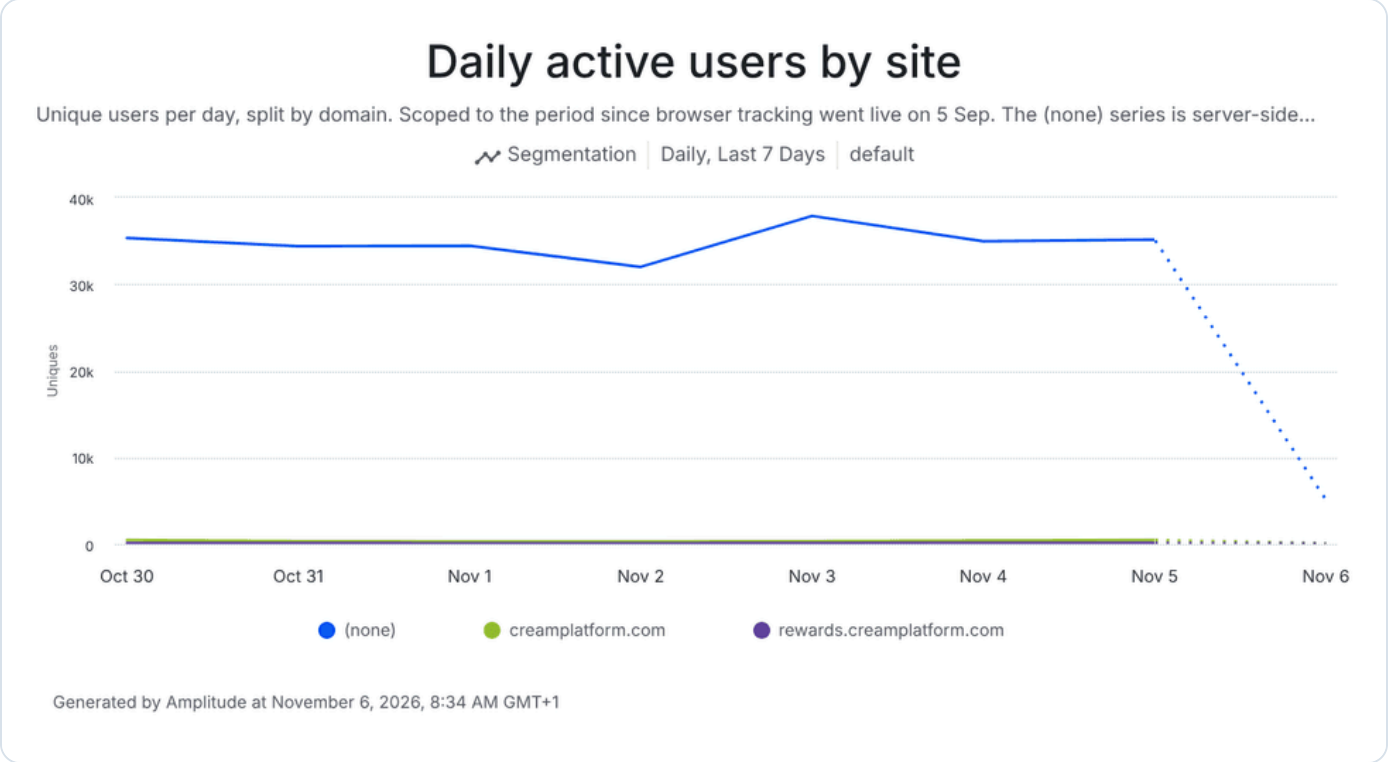


pay.creamplatform.com: CREAM's own wallet for payouts.

RESULTS

What changed for the business

MEASURE	BEFORE	AFTER
Outages	1 to 2 every week	None
Interruption when releasing an update	Site often broke during changes	Under half a second, with automatic rollback
USSD song list load time	Over 10 seconds	Instant
Double charges and paid-but-not-delivered orders	Possible	Blocked by design
Ways to pay	Mostly airtime	Airtime, tokens, card, transfer
Payouts to staff, artists and winners	Outside banks, with fees on each transfer	Through CreamPay
Pages Google can index	Content pages not readable	About 2,700 content pages listed
Languages	English	6
Logins across CREAM apps	A separate sign-in for each app	One sign-in for all of them



Daily unique users, 30 October to 5 November 2026, from CREAM's Amplitude analytics: between about 32,000 and 38,000 people a day. The drop on 6 November is the current day, still in progress when the chart was exported.

THIS MONTH'S TUNE-UP

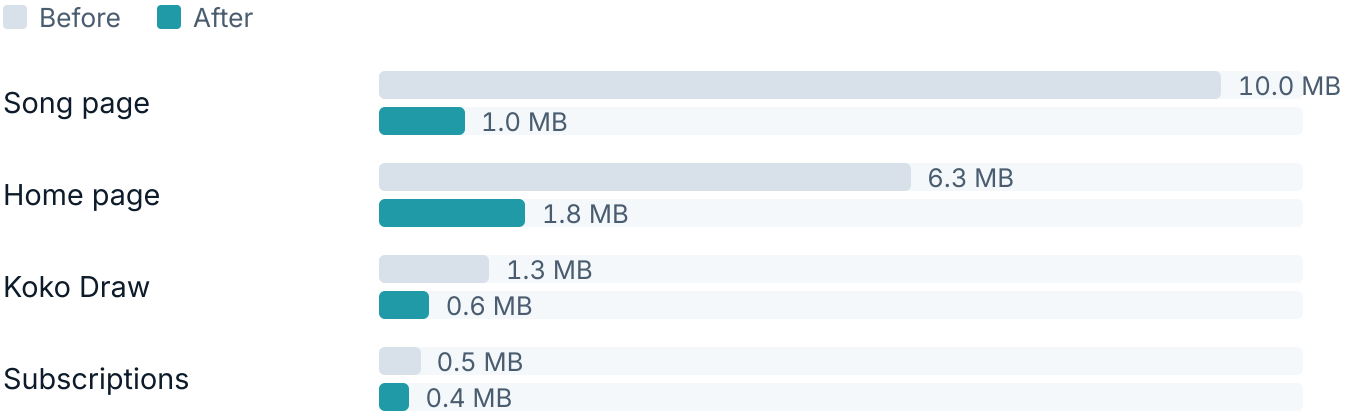
Making the website faster on ordinary phones

Most CREAM fans browse on mid-range Android phones over mobile data. We measured the website the way Google does, using Lighthouse, Google's free website-testing tool, with a simulated mid-range phone on a 4G connection. The tests found three things slowing the site down:

- **Oversized photos.** Small profile pictures were being downloaded at full camera size. One 36-pixel comment avatar was a 6 MB file.
- **Analytics loading first.** The analytics tool, including its session recorder, loaded before the page could respond to taps.
- **Accessibility gaps.** The page blocked pinch-to-zoom, and some headings were out of order for screen readers.

We fixed all three. Photos are now resized to the size they are shown at, analytics loads once the page is ready (still recording everything it did before), and zoom and heading order are fixed. We also flagged a few light grey and yellow text colours that are hard to read, for CREAM's designer to review, since those are brand decisions. For fans who pay

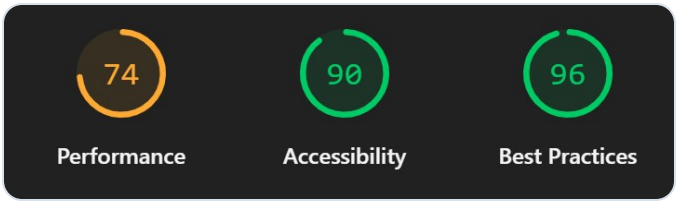
for every megabyte of mobile data, a lighter page is a cheaper page, and a page that opens sooner is a sale that is less likely to be abandoned.



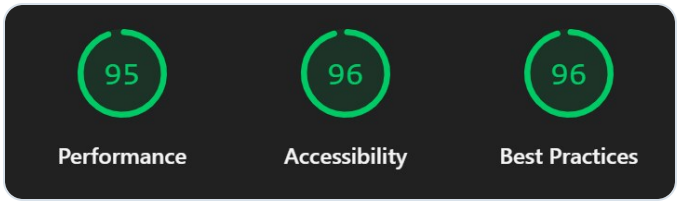
Data a phone downloads to open each page. Lighthouse, simulated mid-range phone on 4G, median of three runs per page, 6 November 2026.

PAGE (ON A PHONE)	READY TO USE BEFORE	READY TO USE AFTER	ACCESSIBILITY BEFORE	AC
Home page	10.4 s	6.6 s	90	
Song page	12.2 s	9.0 s	91	
Koko Draw	8.9 s	5.1 s	93	
Subscriptions	4.2 s	4.0 s	89	

"Ready to use" is the time until the page responds reliably to taps (Lighthouse's Time to Interactive), on a deliberately slowed-down test phone, so real phones on good connections are faster. Accessibility is Lighthouse's score out of 100. Layout shift, which measures content jumping around while it loads, was already 0 and stays 0.



Home page on a laptop, before.



The same page after: performance 74 to 95.

WHAT THIS MEANS FOR YOU

The same approach works for your business

CREAM's problems are common: a system that breaks under load, software that is risky to change, payment routes that give money away, and fees on moving money. We start by finding where your business loses money or customers, then build only what fixes it, and we measure the result.

> Every line we ship is engineered, tested and owned by us. We understand the business first, build what moves it forward, and prove that it worked.

Talk to Kodeguru Tech Solutions

Tell us where your business is losing time or money. We will tell you plainly what technology can fix, what it will cost, and what you should expect to gain.

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